# 12th HACKERS KA JUGAAD

# Magnetism and Matter

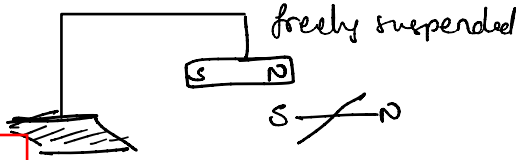
## Magnetism and Matter

A greek philosopher Thales had observed that a naturally occurring ore of iron attracted small pieces of iron towards it. This ore was found in an island in Greece called magnesia. Hence that ore was named magnetite.

## Magnetism

Bits of iron. The phenomena of attraction of small is called, steel,. Cobalt ,nickel etc towards the ore is called Magnetism. The structural formula of this natural ore was Fe3O4

The Chinese discovered that a piece of magnetite when suspended freely, it always points out roughly the north-south direction Thus a natural magnet has attractive and directive properties

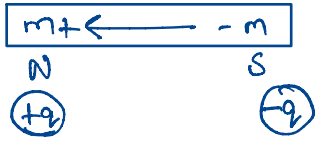


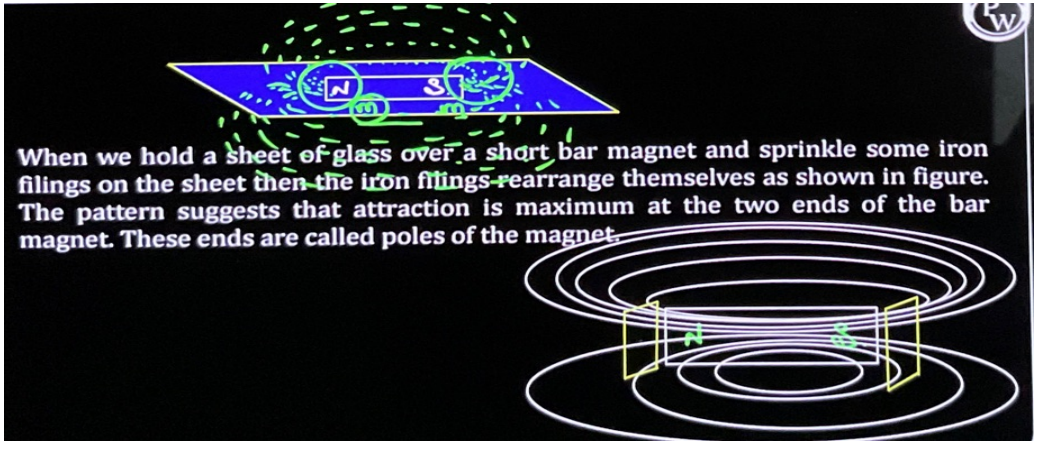

## Artificial Magnets

The natural magnets have often irregular shape and they are weak. It is found that a piece of iron or steel can acquire magnetic properties on rubbing with a magnet. Such magnet made out of iron and steel are called artificial magnets. A bar magnet a horseshoe magnet magnetic needle etc are all artificial magnets.

**Bar Magnet** – It is the most commonly used form of an artificial magnet which have north pole and south pole of same strength at some distance.



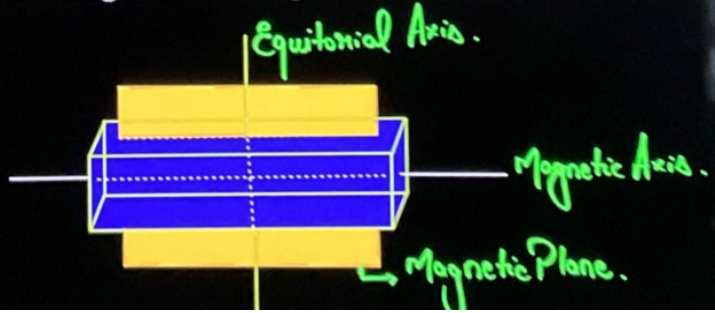




## Basic properties of Bar Magnet

(1) Every magnet attracts small piece of magnetic substances like iron, cobalt, nickel and steel towards it.

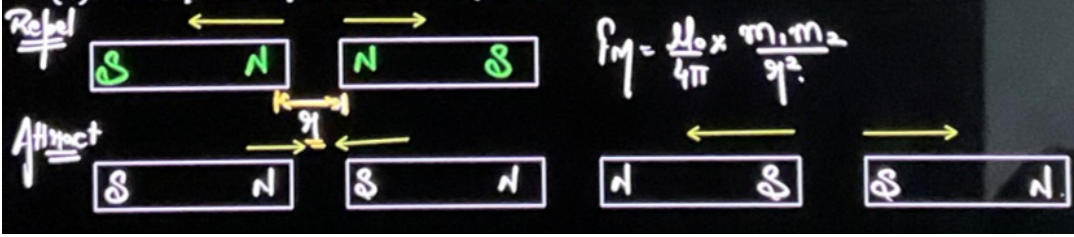
(2) When a magnet is suspended freely with the help of a thread, then it comes to rest along the north south. direction.
The pole of the magnet which points towards geographic north is called north pole and the pole which points towards geographic south is called south pole.

(3) The straight line passing through north and South pole of magnet is called axial line. and the line passing through centre of the magnet in a direction perpendicular to the length of the magnet is called equatorial line of the magnet.




$F_e = \frac{k q_1 q_2}{r^2}$

(4) Like poles repel and unlike poles attract each

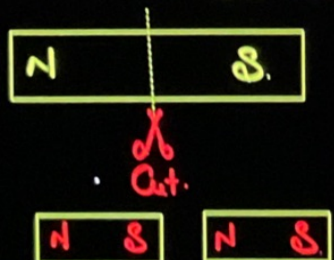



$F_M = \frac{\mu_0}{4\pi} \times \frac{m_1 m_2}{r^2}$

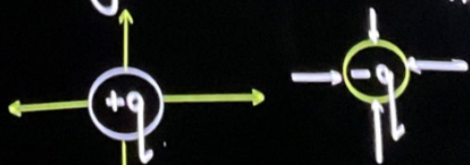
**(5) Inductive property**

when a piece of a magnetic material like soft iron, cobalt, nickel etc. is placed near a bar magnet, it acquires magnetism. The magnetism so acquired is called induced magnetism and this property of magnetism is called inductive property.

(6) The magnetic poles always exist in pairs. The poles of a magnet can never be separated i.e. magnetic monopole do not exists.




→ Charge Can Exist in monopole nature.






$\vec{M}$

m = pole strength

M = Dipole Moment / Magnetic dipole

$\vec{p} = q(2l)$ Dir$^n$ $-q \to q$

$\vec{M} = m(2l)$

↳ sep$^n$ between poles

$-m \longrightarrow +m$ (dir$^n$)

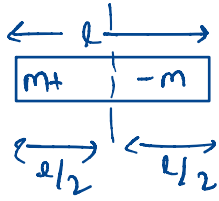
S $\longrightarrow$ N

$(-m)$ $\vec{M}$ $(+m)$

Dipole moment

↳ $\vec{M} = m(2l)$

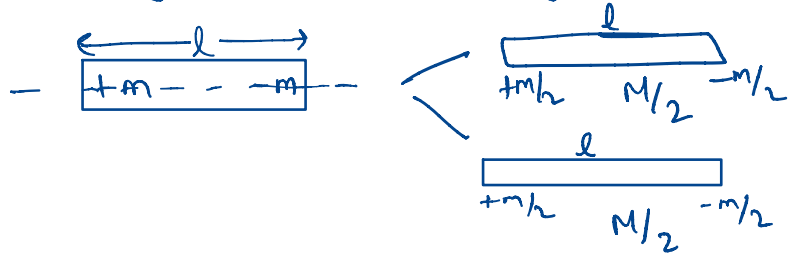
→ cutting of Magnet



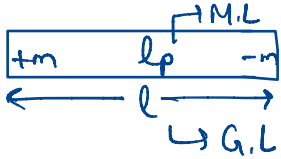

M/2 M/2

If a Bar Magnet is cut ⊥ to the magnetic axis then pole strength remains same but magnetic moment becomes half.

→ cutting of Magnet along the magnetic




when a magnet is cut along the axis then magnetic strength as well as Magnetic Dipole moment becomes half.




$\frac{M.L}{G.L} = 0.84$

## Magnetic Field

It is the space around a magnet or a current carrying conductor in which its magnetic effect can be felt.
In order to visualise a magnetic field graphically, Faraday introduced the concept of the magnetic field lines.

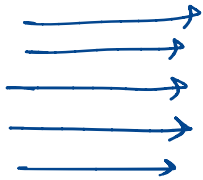
## Magnetic field lines

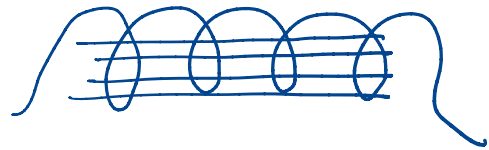
It is an imaginary curve, the tangent to which at any point gives us the direction of magnetic field $\vec{B}$ at that point.




There are basically two types of magnetic field-

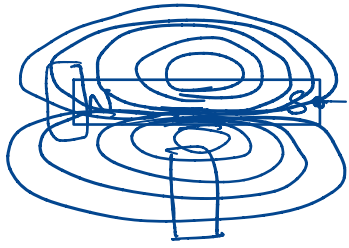
## Uniform Magnetic Field





ex- earth's magnetic field, Magnetic field inside a solenoid carrying current. etc.

## Non-uniform Magnetic Field



no. of field lines ↑.

ex- Magnetic field due to a bar magnet.

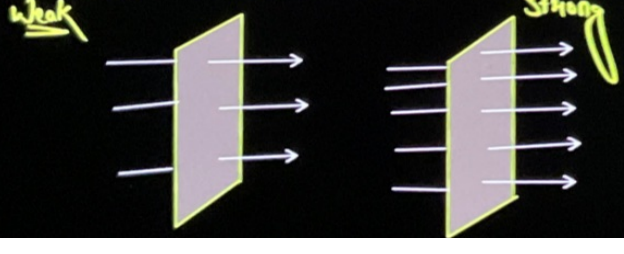
## Properties of magnetic field lines

(1) The magnetic field lines of a magnet (or of a solenoid) form closed continuous loops. The magnetic field lines extending through the body of the magnet/solenoid from south pole to north pole.




(2) Outside the body of the magnet the direction of magnetic field lines is from north pole to south pole.

(3) The magnitude of magnetic field at any point is represented by the number of magnetic field lines passing normally through unit area around that point.





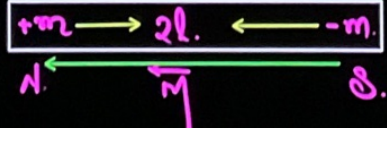
(4) Two magnetic field lines can not intersect each other. Because at the point of intersection we get two directions of magnetic field which is not possible.

(5) Magnetic field lines exists inside every magnetized material.

**Note** - The electric field lines continuous curves which are not closed, whereas magnetic field lines are closed continuous curves.

## Magnetic Dipole

A magnetic dipole consist of two unlike poles of equal strength and separated by a small distance.
The distance between the two poles of a magnetic dipole is called the magnetic length of the magnet. It is a vector directed from south pole to north pole of the magnetic dipole. It is represented by $\vec{2l}$.




## Magnetic Dipole Moment ($\vec{M}$)

It is the product of strength of either pole (m) and the magnetic length ($\vec{2l}$) of the magnet. It's represented by $\vec{M}$.

$$\vec{M} = m(2l)$$

It's a vector quantity and its direction is from south to north pole of the magnet.

- SI unit of M is Joule/Tesla.
- SI unit of pole strength m = A-m

Note:

Magnetic pole strength depend on nature and cross section area of the material

→ when magnet cut into half along the length pole strength becomes half.

→ when magnet cut into half perpendicularly, pole strength remains the same.

## Magnetic field strength due to Bar Magnet

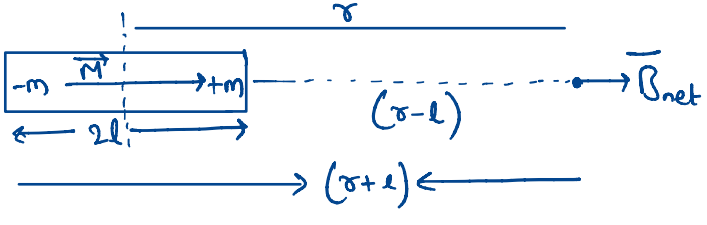
The strength of magnetic field at any point is defined as the force experienced by a hypothetical unit north pole placed at that point.

It is a vector quantity. The direction of magnetic field is along which hypothetical unit north pole would tend to move north pole. would tend to move

(A) When a point lies on axial Line

Let 2*l* be the magnetic length of a bar magnet with centre O. M is the magnetic dipole moment of the magnet.

Where $M = m \times 2l$




$$B_2 = \frac{\mu_0}{4\pi} \times \frac{m}{(r-l)^2} \quad , \quad B_1 = \frac{\mu_0}{4\pi} \times \frac{m}{(r+l)^2}$$

$$B_{net} = B_2 - B_1$$

$$\rightarrow \frac{\mu_0}{4\pi}\times\frac{m}{(r-l)^2} - \frac{\mu_0 m}{4\pi(r+l)^2} \rightarrow \frac{\mu_0 m}{4\pi}\left(\frac{1}{(r-l)^2} - \frac{1}{(r+l)^2}\right)$$

$$B_{net} = \frac{\mu_0 m}{4\pi}\left(\frac{(r+l)^2 - (r-l)^2}{(r^2-l^2)^2}\right)$$

$$B_{net} = \frac{\mu_0 m}{4\pi}\left(\frac{r^2+l^2+2rl - (r^2+l^2-2rl)}{(r^2-l^2)^2}\right)$$

$$\Rightarrow \frac{\mu_0 m \times 2 \times r \times 2l}{(r^2-l^2)^2} \qquad B_{net} = \frac{2\mu_0 r M}{(r^2-l^2)^2},\ l \ll r$$

$$\rightarrow \frac{2\mu_0 \cancel{r} \times M}{r^{\cancel{4}3}} = \boxed{\vec{B}_{net} = \frac{2\mu_0 \vec{M}}{r^3}}$$

If m is the strength of each pole, then magnetic field strength at point P due to s pole of magnet is -

$$B_1 = \frac{\mu_0}{4\pi}\frac{m}{(d+l)^2}$$

Magnetic field strength at P due to N pole of magnet -

$$B_2 = \frac{\mu_0}{4\pi}\frac{m}{(d-l)^2}$$

So the Magnetic field strength at P due to the bar magnet is -

$$B = B_2 - B_1$$

$$B = \frac{\mu_0}{4\pi}\frac{m}{(d-l)^2} - \frac{\mu_0}{4\pi}\times\frac{m}{(d+l)^2}$$

$$= \frac{\mu_0}{4\pi} m\left[\frac{1}{(d-l)^2} - \frac{1}{(d+l)^2}\right]$$

$$= \frac{\mu_0}{4\pi} m\left[\frac{(d+l)^2-(d-l)^2}{(d^2-l^2)^2}\right]$$

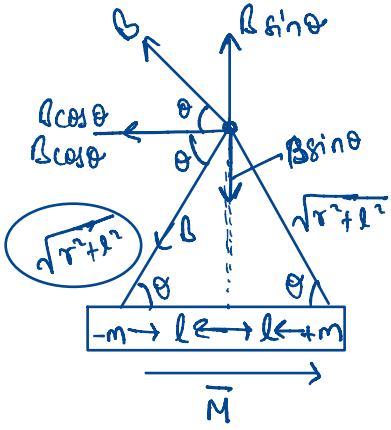
$$B = = \frac{\mu_0}{4\pi} m \cdot \frac{4ld}{(d^2-l^2)^2} = \frac{\mu_0}{4\pi}\frac{2Md}{(d^2-l^2)^2}$$

When the magnet is short $(l^2 \ll d^2)$ so -

$$B = \frac{\mu_0}{4\pi}\frac{2Md}{d^4} \text{ or } \vec{B} = \frac{\mu_0}{4\pi}\frac{2\vec{M}}{d^3} \text{ (Along NP)}$$

At any point on axial line of a dipole, magnetic field strength $\vec{B}$, is along $\vec{M}$

# Magnetic field in equatorial position.




$$\cos\theta = \frac{l}{\sqrt{r^2+l^2}}$$

$$\vec{B}_{net} = -2B\cos\theta.$$

$$\vec{B}_{net} = -2 \times \frac{\mu_0}{4\pi} \times \frac{m}{\sqrt{r^2+l^2}} \propto \frac{l}{\sqrt{r^2+l^2}}$$

$$\Rightarrow -\frac{2\mu_0 m}{4\pi} \times \frac{l}{(r^2+l^2)(r^2+l^2)^{1/2}}$$

$$B_{net} = -\frac{\mu_0 \times m \times 2l}{4\pi (r^2+l^2)^{3/2}}$$

$$B_{net} \Rightarrow \frac{-\mu_0 \vec{M}}{4\pi (r^2+l^2)^{3/2}}$$

$$l << r \rightarrow \frac{-\mu_0 \vec{M}}{4\pi r^2 \times \frac{3}{2}}$$

$$\boxed{\vec{B}_{net} = \frac{-\mu_0 \vec{M}}{4\pi r^3}}$$

## (B) WHEN POINT LIES ON EQUATORIAL LINE

Let here OP = d and point P lies on equatorial line of the bar magnet. Magnetic field strength at P due to North pole of magnet-

$$B_1 = \frac{\mu_0}{4\pi} \frac{m}{(d^2 + l^2)}$$

Magnetic field strength at $P$ due to south pole of magnet -

$$B_2 = \frac{\mu_0}{4\pi} \frac{m}{(d^2 + l^2)}$$

As $B_1 = B_2$ in magnitude, so their vertical components $B_1 \sin\theta$ and $B_2 \sin\theta$ cance each other. However, components along $px$ add up.

Therefore magnetic field strength at $P$ due to the bar magnet is bar magnet is –

$B = B_1\cos\theta + B_2\cos\theta$

$= 2B_1\cos\theta$ (along PX)

so $B = \frac{2\mu_0}{4\pi}\frac{m}{(d^2+\lambda^2)} \times \frac{L}{\sqrt{(d^2+l^2)}}$ $\left[\cos\theta = \frac{L}{\sqrt{d^2+l^2}}\right.$

$$B = \frac{\mu_0}{4\pi}\frac{M}{(d^2+l^2)^{3/2}}$$

If the magnet is short $(l^2 \ll d^2)$ then

$$B = \frac{\mu_0}{4\pi}\frac{M}{d^3} \text{ or } \vec{B} = -\frac{\mu_0}{4\pi}\frac{\vec{M}}{d^3}$$

An any point on equatorial line, magnetic field strength is in the direction opposite to $\vec{M}$.

**Note –**

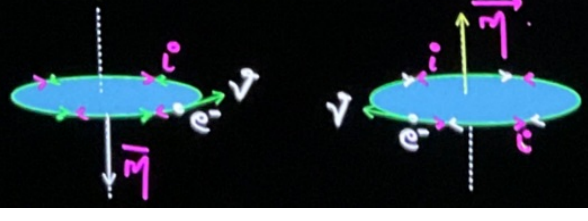
Magnetic field due to a short magnet at any point on axial line of magnet is twice the magnetic field at a point at the same distance on the equatorial line of the magnet.

## Current loop as Magnetic Dipole

Consider a plane loop of wire carrying current in anticlockwise direction. Therefore, it has a north polarity.

Looking at the lower face of the loop current is clockwise, therefore it has a south polarity

$\vec{m} = i\vec{A}$.



This the current carrying loop behaves as a magnetic dipole.
The magnetic dipole moment of the current loop (M) is directly proportional to

$$M \propto I$$
$$\& M \propto A$$

So

$$\begin{cases} M = KIA \\ M = IA \end{cases}$$

for N turns $\vec{M} = NI\vec{A}$

SI Unit - ampere meter$^2$ ($Am^2$)

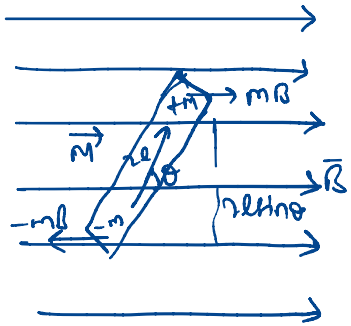
**1 ampere meter$^2$ -**

It is the magnetic moment of one turn loop of area one square meter carrying a current of one ampere.

# Torque on a Bar Magnet in a magnetic field

The magnet is held at angle θ with the direction of $\vec{B}$.
Force on north pole = mB ( along $\vec{B}$)
Force on south pole = mB (opposite to $\vec{B}$ )
These forces, are equal and tends to rotate the magnet clockwise.
Torque acting on the bars magnet is -

$$\text{Torque } (\tau) = \text{Force} \times \text{perpendicular distance}$$
$$= mB \times NA$$




$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$|\tau| = F \times r_\perp$$
$$\rightarrow MB \times 2l \sin\theta$$
$$|\tau| = 2mlB\sin\theta$$
$$|\vec{\tau}| = MB\sin\theta$$
$$|\vec{\tau}| = \vec{M} \times \vec{B}$$

when $B = 1$ and $\theta = 90°$ then

$$\tau = M \times 1 \times \sin 90° \Rightarrow \tau = M$$

Magnetic dipole moment is the torque acting on a dipole held perpendicular to a uniform magnetic field of unit strength.

SI Unit of $\vec{M}$ = Joule per Tesla ($JT^{-1}$)

# Potential Energy of a Magnetic Dipole in a magnetic field.

The energy possessed by the dipole due to its particular position in the field is the potential energy of a magnetic dipole.

When a magnetic dipole of moment $\vec{M}$ is held at an angle $\theta$ with the direction of a uniform magnetic field $\vec{B}$, So the magnitude of the torque acting on the dipole is -

$$\tau = MB\sin\theta$$

$W = \int_{\theta_1}^{\theta_2} MB\sin\theta \, d\theta$

$= MB(-\cos\theta)_{\theta_1}^{\theta_2}$

$W = -MB(\cos\theta_2 - \cos\theta_1)$

$W = \int \vec{F}\cdot d\vec{s}, \quad \int \tau \, d\theta$

$\theta_1 = \pi/2$

$U = -MB\cos\theta$

This torque tend to align the dipole in the direction of the field. Work has to be done in rotating the dipole against the action of the torque. This work done is stored in the magnetic dipole as potential energy of dipole.

Now, small amount of work done in rotating the dipole through a small angle do against the rotating torque is -

$d\omega = \tau d\theta = MB\sin\theta d\theta$

Total work done in rotating the dipole from $\theta_1$ to $\theta_2$ is

$$w = \int_{\theta_1}^{\theta_2} MB\sin\theta d\theta = MB[-\cos\theta]_{\theta_1}^{\theta_2}$$

$$w = -MB[\cos\theta_2 - \cos\theta_1]$$

So the potential energy of the dipole is -

$$\Delta U = \omega = -MB(\cos\theta_2 - \cos\theta_1)$$

When $\theta_1 = 90°$ and $\theta_2 = \theta$ then.

$$U = W = -MB\cos\theta$$

$$U = -\vec{M}\cdot\vec{B}$$

## SPECIAL CASES

(i) when $\theta = 90°$ then $U = -MB\cos 90° = 0$

when the dipole is perpendicular to magnetic field its potential energy is zero.

(ii) when $\theta = 0°$ then $U = -MB\cos 0° = -MB$ Stable.

If is minimum. This is the position of stable equilibrium.

so when the magnetic dipole is aligned along the magnetic field, it is in stable equilibrium having minimum P.E.

(iii) when $\theta = 180°$ then $U = -MB\cos 180° = MB$ "Unstable"

It is maximum. This is the position of unstable equilibrium.

## SOME IMPORTANT TERMS USED IN MAGNETISM

### (i) Magnetic Permeability ($\mu$).

It is the ability of a material to permit the passes of magnetic lines of force through it. The degree or extent to which magnetic fists can penetrate or permeate a material is called relative magnetic permeability of the material. It is represented by $\mu_r$.

$\mu_0$ = Permeability in Air.

$\mu_r$ = Relative Permiability.

$\mu_m$ = Permeability in Medium

**Relative Magnetic permeability of a material**

The relative magnetic permeability of a material may also be defined as the ratio of magnetic permeability of the material ($\mu$) did magnetic permeability of free space ($\mu_0$).

$\mu_r = \frac{\mu}{\mu_0}$ $\quad \mu_r = \frac{\mu_m}{\mu_0}$ $\quad \mu_r = \frac{B_m}{B_0}$

or $\mu_r = \frac{B}{B_0}$

$\mu_r$ has no dimensions. Its value is 1 for vacuum.

$$\mu_0 = 4\pi \times 10^{-7} \omega b A^{-1} m^{-1}$$

### (ii) Magnetising Force or Magnetic Intensity (H)

The degree to which a magnetic field can magnetise a material is represented in terms of magnetising force or magnetic intensity ($H$).

Lets consider a toroidal solenoid with $n$ turns per unit length carrying a current I wound round a ring of a magnetic material.

The magnetic field produced in the material is

$$B = \mu n I$$

The product $nI$ is called the magnetising force or magnetic intensity $H$.

so $B = \mu H$ $\quad \uparrow \mu = \frac{B}{H}$, $\mu = \frac{B_m}{H_m}$.

or $H = \frac{B}{\mu}$

If inside the toroidal solenoid, there is free space, then magnetic field.

$$B_0 = \mu_0 H$$

or $H = \frac{B_0}{\mu_0}$

$I = \frac{B_m}{H_0}$

The SI unit of H is Ampere turn/m($Am^{-1}$)

**(iii) Magnetisation Intensity (I)**

The Magnetisation intensity of a magnetic material is defined as the magnetic moment per unit volume of the material.

$$I = \frac{\text{Magnetic moment}}{\text{volume}} = \frac{M}{V}$$

$$I = \frac{m \times 2L}{A \times 2L} = \frac{m}{A} \quad I = \frac{B_m}{\mu_0}$$

Hence the Magnetisation intensity of a material is also defined as the pole strength per unit area of. cross-section of the material.

$$I = \frac{\text{magnetic moment}}{\text{Volume}} = \frac{\text{Amp} \cdot \text{metre}}{\text{metre}} = Am^{-1}$$

So the SI unit of I is A $m^{-1}$

**(iv) Magnetic Susceptibility ($\chi_m$)**

It is a property which determine how easily a material can be magnetised. susceptibility of a magnetic material is defined as the ratio of the intensity of magnetisation (I) induced in the material to the magnetising force (H) applied on it.

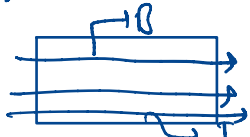
Magnetic susceptibility is represented by $\chi_m$.

$$\chi_m = \frac{I}{H}$$

As unit of $I$ and $H$ are same $(Am^{-1})$, therefore $(\chi_m)$ has no units and no dimensions.

## Relation between magnetic Permeability and magnetic Susceptibility

when a magnetic material is placed in a magnetic field of magnetic intensity H, the material gets magnetised.



$\rightarrow B = B_0 + B_m$

$B_{net} = \mu H = \mu_0 H + \mu_0 I$

$\mu H = \mu_0 (H + I)$

$\frac{\mu}{\mu_0} = I + \frac{I}{H}$ , $\mu_m = 1 + \chi_m$

The total magnetic field $B$ in the material is the sum of the magnetic field $B_0$ in vacuum produced by the magnetic intensity and, magnetic field $B_m$, due to magnetisation of the material.

$$B = B_0 + B_m$$

But $B_0 = \mu_0 H$ $B_m = \mu_0 I$ and $B = \mu_H$

So $\mu_H = \mu_0 H + \mu_0 I$

dividing by $\mu_0 H$ we get

$$\frac{\mu}{\mu_0} = 1 + \frac{I}{H}$$

So

$$\mu_r = 1 + \chi_m$$

as $\chi_m = I/H$

This is the relation between the relative magnetic permeability and magnetic susceptibility of the material.

## Classification of Magnetic Materials

Diamagnetic | Paramagnetic | Ferromagnetic

### i) Diamagnetic substances.

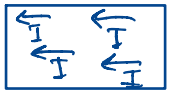
The diamagnetic substances are those in which the individual atoms or molecules or ions do not possess any of net magnetic moment on their own

When such substances are placed in an external magnetic field, they get weekly magnetised in a direction opposite to the magnetising field

Examples, antimony, bismuth, copper, lead, quartz, water, mercury hydrogen, inert gases etc

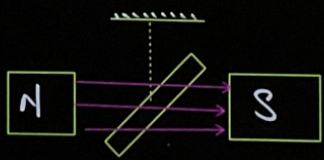


$\vec{M}_{net} = 0$

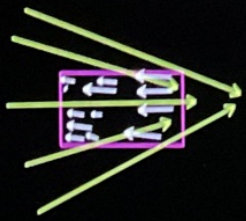


# Properties

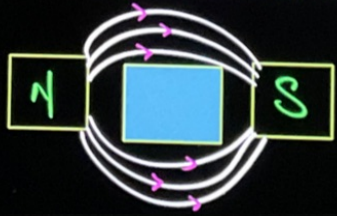
(i) When diamagnetic substances are suspended in a uniform magnetic field, they set their longest axis at right angle to the direction of the field.




(ii) When diamagnetic substances are placed in a nonuniform magnetic field, these substances have a tendency to move from stronger parts of the field to the weaker parts.



(iii) When a piece of diamagnetic material is placed in a magnetising field, the magnetic field lines prefer not to pass through the material.
It implies that the magnetic field lines are repelled and the field inside the material is reduced slightly.




(iv) Relative magnetic permeability of diamagnetic substances is always less than unity.

$$\mu_r = \frac{B}{B_0} \text{ as } B < B_0$$

so $\mu_r < 1$

As $\mu_r = 1 + \chi_m$

(v) and $\mu_r < 1$ so $\chi_m$ is negative
Hence susceptibility of diamagnetic material has a small negative value.

(vi) Susceptibility of diamagretics does not change with temperature.

# Explanation of Diamagnetism

In a diamagnetic substance, the motion of all the electrons in an atom is assumed as the motion of two electrons revolving with same angular velocity in the circular orbit of same radius but in opposite sense

In the absence of any external magnetic field, the magnetic moment of two electrons be equal and opposite and cancel each other for every atom so that net magnetic moment of each item is zero

When a uniform magnetic field B be applied then each electron experiences a magnetic Lorentz force F equal to eVA. According to Flemings left hand rule the Lorentz force F acts radially out wards on the electron revolving anticlockwise. it tends to decrease the centripetal force and hence decrease the velocity of electron to $(\vec{v} - \Delta\vec{v})$ As a result. The magnetic moment of electron decreases to $(\vec{M} - \Delta\vec{M})$.

On the electron revolving clockwise, the Lorentz force acts radially inwards, tending to increase the centripetal force and hence the velocity of electron to $(\vec{v} + \Delta\vec{v})$. Therefore, the magnetic moment of electron increases to $(\vec{M} + \Delta\vec{M})$.

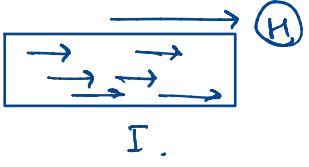
The vector addition of these two magnetic moments gives rise to a net dipde magnetic moment $2\Delta\vec{M}$, directed upward i.e. opposite to external field $\vec{B}$.
The net magnetic moment $2\Delta\vec{M}$ so developed is called induced magnetic moment. The magnetic moment induced in different atoms add vertically to give a net magnetization of the material in a direction opposite to $\vec{B}$, This accounts for the diamagnetic behavior of material.

* The appearance of induced magnetic moment in atoms is not affected by the thermal motion of atoms, therefore, magnetic susceptibility of such materials does not depends on temperature of the materials.

## ii) Paramagnetic Substances -

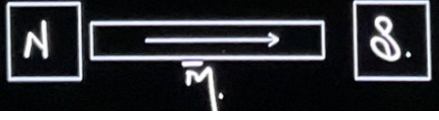
Paramagnetic substances are those in which each individual atom or molecule or iron has a net non zero magnetic moment of its own. When such substances are placed in an external field they get weekly Magnetised in the direction of the magnetising field.

Examples, aluminium, chromium, manganese, lithium, platinum, tungsten, sodium, oxygen ,etc



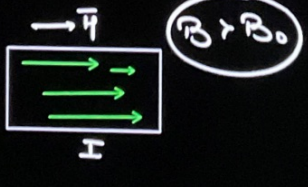

### Properties

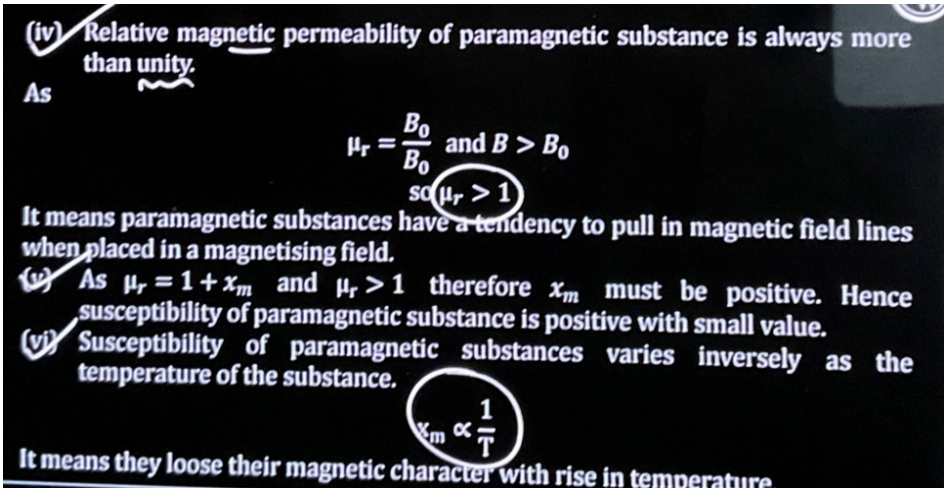
(i) When a paramagnetic substance is suspended in a uniform magnetic field, they rotate so as to bring their longest axis along the direction of the field.




(ii) When the paramagnetic materials are placed in. a non-uniform magnetic field they tend to move from weaker part's of the field to the stronger parts. It implies that paramagnetic substances get weakly. attracted to the magnet.

(iii) When a piece of a paramagnetic substance is placed in a magnetising field, the magnetic field lines prefer to pass through the piece rather than through Thus magnetic field B inside the piece is more than the magnetic field $B_0$ outside the piece. $B > B_0$.






## Explanation of Paramagnetism

In paramagnetic materials, every atom has some permanent magnetic dipole moment. In the absence of external magnetic field. The atomic dipole are randomly oriented so that Average magnetic moment per unit volume of the material is zero. Therefore, a paramagnetic material does not behave as a magnet in the absence of an external magnetic field

When an external magnetic field is applied, it tries to align the atomic magnetic dipoles in the direction of the field. That is why the material gets magnetised weakly in the direction of the field. This is ParaMagnetism

When we raise the temperature of the material, the atomic magnetic dipoles acquires of kinetic energy. This tend to disorient the dipoles that is why magnetic susceptibility of the paramagnetic materials decreases with rise in temperature.

## (iii) Ferromagnetic Substances

Ferromagnetic substances are those in which each individual atom/molecule/ion has a non zero magnetic moment, as in a paramagnetic substances. when such substances are placed in an external magnetising field, they get strongly magnetised in the direction of the field.

**Examples** - Iron, Cobalt, Nickel and a number of their alloys.

## Properties

(i) They are strongly magnetised in the direction of external field in which they are placed-

(ii) They have a strong tendency to move from a region of weak magnetic field to the region of strong magnetic field-Hence they get strongly attracted to a magnet.

(iii) Relative magnetic permeability of feromagnetic materials is very large. $(\approx 10^3 \text{ to } 10^5)$

$$\mu_R \gg 1$$

(iv) The susceptibility of ferromagnetic materials is very, large. That is why they can be magnetised easily and strongly.

$$x_m = \mu_r - 1 \qquad x_m = \text{large positive value}$$

(v) With rise in temperature, susceptibility of ferromagnetic materials decreases. At a certain temperature ferromagnetic change over to paramagnetic. This transition temperature is called curie temperature. For example, curie temperature of iron is about 1000 K.

(vi) When a piece of a ferromagnetic substance is placed in a magnetising field, the magnetic field lines prefer to pass through the piece rather than through air:

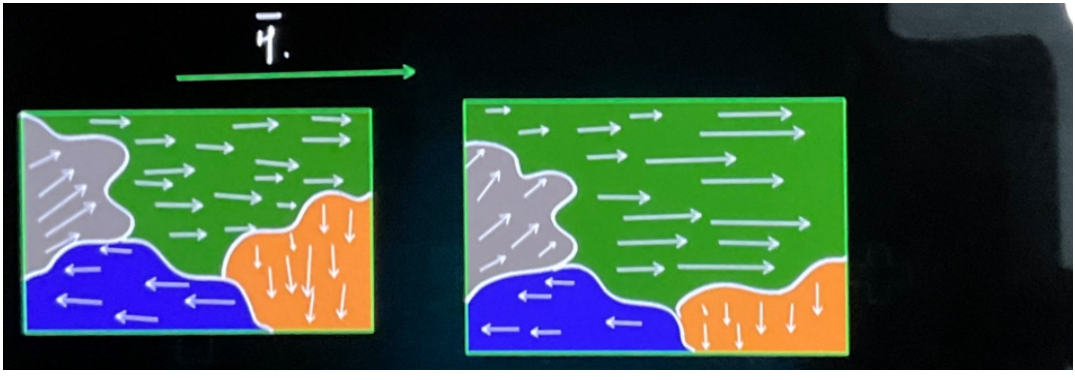


**Explanation:-**

Each atom of ferromagnetic substance is a tiny magnetic dipole having permanent dipole moment.

In ferromagnetic substances, atoms form a very large number of small effective regions called domains. Each domain is a strong magnet without any external magnetic field.

Inspite of this, a ferromagnetic substance does not behave as a magnet, because in the absence. of external magnetic field the magnetic moments of different domains are randomly oriented so that their resultant magnetic moment in any direction is zero.
when an external magnetic field is applied on the ferromagnetic substance it gets strongly magnetised.




The domains starts to rotate till their magnetic moments are aligned in the direction of the applied magnetic field. and arranged to form a single giant domain. This is how ferromagnetic of the applied field.

**Soft Ferromagnet :-**
when external magnetic field is removed, magnetisation disappears in some of the materials like soft iron. These materials are called soft ferromagnets.

**Hard Ferromagnet:-**
In materials like Alnico (an alloy of iron, aluminium, nickel, cobalt and copper) the magnetisation persists even after removal of external magnetic field. Such materials are called hard ferromagnets?

- When ferromagnetic materials is heated, its magnetisation decreases gradually. This is because the domain structure disturbs with rise in temperature

| Property | Diamagnetic substances | Paramagnetic substances | Ferromagnetic substance |
|---|---|---|---|
| Effect of magnets | They are feebly repelled by magnets. | They are feebly attracted by magnets. | They are strongly attracted by magnets |
| In external magnetic field | Acquire feeble magnetisation in the opposite direction of the magnetising field. | Acquire feeble magnetisation in the direction of the magnetising field | Acquire strong magnetisation in the direction of the magnetising field |
| In a non-uniform magnetic field | Tend to move slowly from stronger to weaker parts of the field. | Tend to move slowly from weaker to stronger parts of the field. | Tend to move quickly from weaker to stronger parts of the field. |

| Property | Diamagnetic substances | Paramagnetic substances | Ferromagnetic substance |
|---|---|---|---|
| In a uniform magnetic field | A freely suspended diamagnetic rod aligns itself perpendicular to the field | A freely suspended diamagnetic rod aligns itself perpendicular to the field. | A freely suspended ferromagnetic rod aligns itself parallel to the field |
| Susceptibility value ( $\chi_m$) | Susceptibility is small and negative. $-1 \leq \chi_m > 0$ | Susceptibility is small and positive, $0 < \chi_m < \varepsilon$, where $\varepsilon$ is a small number | Susceptibility is very large and positive $\chi_m > 1000$ |
| Relative permeability value ($\mu_r$) | Slightly less than 1 $0 \leq \mu_r < 1$ | Slightly greater than 1 $1 < \mu, < 1 + \varepsilon$ | Of the order of thousands $\mu_r > 1000$ |

| Property | Diamagnetic substances | Paramagnetic substances | Ferromagnetic substance |
|---|---|---|---|
| Permeability value ($\mu$) | $\mu < \mu_0$ | $\mu > \mu_0$ | $\mu >> \mu_0$ |
| Effect of temperature | Susceptibility is independent of temperature. | Susceptibility varies inversely as temperature : $\chi_m \propto \frac{f}{T}$ | Susceptibility decreases with temperature in a complex manner. $\chi_m \propto \frac{1}{T - T_C} (T > T_C)$ |
| Removal of magnetizing field | Magnetisation lasts as long as the magnetising field is applied. | As soon as the magnetizing field is removed, magnetization is last. | Magnetisation is retained even after the magnetising field removed. |

| Property | Diamagnetic substances | Paramagnetic substances | Ferromagnetic substance |
|---|---|---|---|
| Variation of M with H | M changes linearly with H. | M changes linearly with H and attains saturation at low temperature and in very strong fields. | M changes with H non-linearly and ultimately attains saturation. |
| Hysteresis effect | B-vector shows no hysteresis | B-vector shows no hysteresis | B-vector shows hysteresis. |
| Physical state of the material | Solid, liquid of gas. | Solid, liquid or gas. | Normally solids only. |
| Examples | Bi, Cu, Pb, Si, $N_2$ (at STP), $H_2O$, NaCl | Al, Na, Ca, $O_2$ (at STP),$CuCl_2$ | Fe, Ni, Co, Gd, $Fe_2O_3$, alnico. |

THANK YOU !!